

CHEMAL, Russian Federation

WMO: 360580

Lat: 51.388N

Lon: 86.021E

Elev: 419

StdP: 96.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.6	-26.2	-31.6	0.2	-29.5	-28.1	0.3	-25.9	10.2	-7.5	8.5	-8.2	0.1	340	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	31.2	19.0	29.2	18.4	27.5	17.6	20.4	28.4	19.4	27.0	18.6	25.4	1.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.5	22.8	16.9	12.7	22.0	16.1	12.0	21.1	60.9	28.4	57.4	27.0	54.5	25.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.4	3.9	-31.9	35.5	3.7	1.9	-34.5	36.9	-36.7	38.1	-38.7	39.1	-41.4	40.6		
(5)			-31.8	22.2	3.6	1.4	-34.4	23.2	-36.4	24.0	-38.5	24.8	-41.0	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-12.0	-8.8	-0.5	7.0	12.6	18.0	19.6	17.7	12.1	5.9	-2.5	-9.1
	DBStd	12.14	6.99	6.92	6.13	5.23	4.62	3.56	2.88	3.47	4.24	4.45	6.81	6.72
	HDD10.0	2814	682	526	327	114	28	1	0	1	26	139	377	593
	HDD18.3	4979	941	759	584	339	185	48	19	52	190	384	626	852
	CDD10.0	1017	0	0	1	25	108	241	299	240	89	13	1	0
	CDD18.3	141	0	0	0	0	6	38	60	34	3	0	0	0
	CDH23.3	1747	0	0	0	28	181	473	583	407	74	1	0	0
	CDH26.7	554	0	0	0	5	51	149	197	135	17	0	0	0
Wind	WSAvg	1.3	1.9	1.3	1.1	1.2	1.3	1.2	1.0	1.0	1.1	1.3	1.5	1.9
Precipitation	PrecAvg	540	10	10	17	41	66	87	105	88	51	30	24	12
	PrecMax	673	20	23	37	96	161	144	155	150	119	77	65	30
	PrecMin	381	3	1	4	11	31	37	69	45	17	11	6	1
	PrecStd	71	5	6	9	22	29	26	24	29	26	17	15	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	9.0	18.3	27.0	31.2	32.4	34.0	33.9	29.1	21.2	15.2	8.9
		MCWB	2.7	3.4	8.6	13.1	17.4	18.8	20.5	20.3	16.4	11.5	7.7	3.2
	2%	DB	3.6	5.9	14.5	22.7	27.8	30.3	30.9	30.2	25.3	18.3	12.1	5.4
		MCWB	-0.5	1.1	6.5	11.3	15.7	18.4	19.6	19.1	15.1	10.2	6.0	0.5
	5%	DB	0.7	3.3	11.7	19.7	25.2	28.3	28.8	27.7	22.5	15.8	9.1	3.1
		MCWB	-2.6	-0.7	5.2	10.1	14.1	17.8	19.1	18.2	13.9	9.0	3.9	-0.9
	10%	DB	-2.1	0.7	8.7	16.9	22.2	26.1	26.7	25.5	19.8	13.3	6.0	0.5
		MCWB	-4.5	-2.4	3.3	8.7	13.0	17.1	18.2	17.2	12.8	7.5	2.1	-2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.6	9.0	13.8	19.0	20.9	22.8	21.5	17.6	12.3	8.2	3.1
		MCDB	7.8	8.5	17.0	24.0	29.3	29.2	31.0	30.5	27.0	19.0	14.7	8.5
	2%	WB	-0.2	1.3	7.0	12.2	16.6	19.6	20.7	19.9	15.9	10.7	6.1	0.7
		MCDB	3.2	5.4	13.7	20.9	25.1	27.6	28.2	28.3	23.7	17.0	11.5	4.8
	5%	WB	-2.5	-0.6	5.4	10.7	15.0	18.6	19.7	18.8	14.6	9.3	4.2	-0.8
		MCDB	0.4	2.9	11.2	18.1	23.1	26.1	26.9	26.1	21.1	15.0	8.4	2.9
	10%	WB	-4.5	-2.3	3.6	9.3	13.7	17.7	18.8	17.8	13.4	7.9	2.4	-2.4
		MCDB	-2.0	0.6	8.3	16.0	20.5	24.5	25.1	23.9	18.9	12.8	5.8	0.4
Mean Daily Temperature Range	5% DB	MDBR	9.6	13.0	14.3	13.9	14.1	13.0	12.2	12.6	12.8	11.0	8.9	8.1
		MCDBR	10.3	13.2	17.2	20.0	19.9	17.6	16.5	17.6	18.0	15.6	12.2	10.2
		MCWBR	7.1	9.2	9.9	10.1	9.3	7.3	6.5	7.4	8.2	8.3	7.2	6.9
	5% WB	MCDBR	10.2	13.1	16.2	17.9	17.1	15.0	14.6	15.7	15.9	14.7	11.6	9.9
		MCWBR	7.2	9.2	9.5	9.5	8.4	6.8	6.5	7.0	7.5	8.0	7.1	6.8
Clear-Sky Solar Irradiance	taub	0.290	0.327	0.388	0.450	0.437	0.432	0.449	0.425	0.380	0.366	0.324	0.293	
	taud	2.073	2.009	1.942	1.964	2.098	2.210	2.181	2.224	2.286	2.191	2.110	2.041	
	Ebn at Noon	689	762	782	781	818	826	802	802	798	717	635	604	
	Edh at Noon	80	113	148	164	150	136	138	126	105	95	77	70	
All-Sky Solar Radiation	RadAvg	1.33	2.40	3.83	4.61	5.28	5.84	5.35	4.78	3.58	2.06	1.21	0.96	
	RadStd	0.12	0.15	0.30	0.35	0.51	0.54	0.32	0.35	0.30	0.24	0.11	0.10	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page